

Derivation of the Gain Measurement a_{xm} and its Noise Statistics ($\delta x_m \rightarrow a_{xm}$)

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k - d] + n_x[k], \quad d = 2,$$

where $\delta x[k] = x_d[k] - x[k]$ is the tracking error, $n_x[k] \sim \mathcal{N}(0, \sigma_n^2)$ is the sensor noise, and λ_c ($0 \leq \lambda_c < 1$) is the closed-loop pole.

Closed-loop error

Control objective: $\delta x[k + 1] = \lambda_c \delta x[k]$.

Control law (4-state; the gain rate is fed forward, the disturbance is not compensated):

$$f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k - i] f_{dx}[k - i] \right] \right\},$$

with $\Delta x_d^d[k] = \Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k - i]$ the (delay-compensated) desired-trajectory feedforward. Under this law, and dropping the gain-estimation and disturbance terms (which are higher order), the closed-loop error becomes

$$\delta x[k + 1] = \lambda_c \delta x[k] - \delta x_T^d[k] - (1 - \lambda_c) n_x[k],$$

$$\delta x_T^d[k] = \delta x_T[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_T[k - i], \quad \delta x_T[k] := a_x[k] f_T[k] \quad (\text{white, } \sigma_{\delta x_T}^2 = 4k_B T a_x).$$

Solving this recursion in the z -domain gives two noise channels:

$$\delta x(z^{-1}) = -H_T(z^{-1}) \delta x_T(z^{-1}) - H_n(z^{-1}) n_x(z^{-1}),$$

$$H_T(z^{-1}) = z^{-1} \frac{1 + (1 - \lambda_c)z^{-1} + (1 - \lambda_c)z^{-2}}{1 - \lambda_c z^{-1}}, \quad H_n(z^{-1}) = z^{-1} \frac{1 - \lambda_c}{1 - \lambda_c z^{-1}}.$$

IIR mean removal

Low-pass (EWMA) mean and high-pass residual:

$$\delta x_{md}[k] = (1 - a_{pd}) \delta x_{md}[k - 1] + a_{pd} \delta x_m[k], \quad H_{LP}(z^{-1}) = \frac{a_{pd}}{1 - (1 - a_{pd})z^{-1}},$$

$$\delta x_r[k] = \delta x_m[k] - \delta x_{md}[k], \quad H_{HP}(z^{-1}) = 1 - H_{LP}(z^{-1}) = \frac{(1 - a_{pd})(1 - z^{-1})}{1 - (1 - a_{pd})z^{-1}}.$$

Since $H_{HP}(z=1) = 0$, the residual δx_r is exactly zero-mean.

Channel decomposition (substitute $\delta x_m = \delta x[k - d] + n_x$, independent $\delta x_T \perp n_x$):

$$\delta x_r(z^{-1}) = -F_T(z^{-1}) \delta x_T(z^{-1}) + F_N(z^{-1}) n_x(z^{-1})$$

$$F_T = H_{HP} z^{-d} H_T, \quad F_N = H_{HP} [1 - z^{-d} H_n] = (1 - z^{-1}) z^3 F_T.$$

Residual variance (by Parseval, residues at the poles $z = 1 - a_{pd}$ and $z = \lambda_c$; $D := 1 - (1 - a_{pd})\lambda_c$):

$$\boxed{\sigma_{\delta x_r}^2 = C_{\text{dpmr}}(\lambda_c, a_{pd}) \sigma_{\delta x_T}^2 + C_n(\lambda_c, a_{pd}) \sigma_n^2}$$

$$C_{\text{dpmr}} = (1 - a_{pd})^2 \left[\frac{2(1 - a_{pd})(1 - \lambda_c)}{D} + \frac{2}{2 - a_{pd}} \cdot \frac{1}{(1 + \lambda_c) D} \right],$$

$$C_n = \frac{2(1 - a_{pd})^2}{2 - a_{pd}} \left[1 + \frac{(1 - a_{pd})^2 a_{pd}(1 - \lambda_c)}{D} + \frac{(1 - \lambda_c)^2}{(1 + \lambda_c) D} \right] \quad (\text{full } a_{pd} \text{ closed form}).$$

Variance estimate and gain measurement

EWMA variance of the residual:

$$\hat{\sigma}_{\delta x_r}^2[k] = (1 - a_{cov}) \hat{\sigma}_{\delta x_r}^2[k - 1] + a_{cov} \delta x_r^2[k], \quad H_E(z^{-1}) = \frac{a_{cov}}{1 - (1 - a_{cov})z^{-1}}.$$

Because $H_{HP}(z=1) = 0$ (zero-mean residual) and $H_E(z=1) = 1$, the estimate is unbiased:

$$\mathbb{E}[\hat{\sigma}_{\delta x_r}^2] = \sigma_{\delta x_r}^2 = C_{\text{dpmr}} 4k_B T a_x + C_n \sigma_n^2.$$

Linear inversion (linear in $\hat{\sigma}_{\delta x_r}^2$, not a square root — the variance is itself *linear* in a_x):

$$\boxed{a_{xm}[k] = \frac{\hat{\sigma}_{\delta x_r}^2[k] - C_n \sigma_n^2}{C_{\text{dpmr}} \cdot 4k_B T}, \quad \mathbb{E}[a_{xm}] = a_x.}$$

Measurement noise $R(2, 2) = \text{Var}(a_{xm})$

Variance scaling. a_{xm} is an affine map of $\hat{\sigma}_{\delta x_r}^2$ (slope $m = C_{\text{dpmr}} 4k_B T$, offset $C_n \sigma_n^2$), so the variance scales by $1/m^2$ (this also converts the units, $\mu\text{m}^4 \rightarrow (\mu\text{m}/\text{pN})^2$):

$$R(2, 2) = \text{Var}(a_{xm}) = \frac{\text{Var}(\hat{\sigma}_{\delta x_r}^2)}{(C_{\text{dpmr}} \cdot 4k_B T)^2}.$$

Variance of the EWMA estimate (three independent layers):

$$\text{Var}(\hat{\sigma}_{\delta x_r}^2) = K_{\text{var}} \sigma_{\delta x_r}^4 \text{IF}, \quad K_{\text{var}} = \frac{2a_{cov}}{2 - a_{cov}},$$

where K_{var} combines Isserlis' fourth moment ($\text{Var}(X^2) = 2\sigma^4$ for Gaussian X) with the EWMA variance gain $a_{cov}/(2 - a_{cov})$, and IF is the color-inflation factor: δx_r inherits the poles $1 - a_{pd}$ and λ_c , so it is autocorrelated and so is δx_r^2 ,

$$\text{IF} = 1 + 2 \sum_{\tau \geq 1} \rho_{\delta x_r}^2(\tau) s^\tau, \quad s = 1 - a_{cov}.$$

The square ρ^2 comes from Isserlis at lag τ : $\text{Cov}(\delta x_r^2[k], \delta x_r^2[k + \tau]) = 2 \text{Cov}(\delta x_r[k], \delta x_r[k + \tau])^2 = 2\rho_{\delta x_r}^2(\tau) \sigma_{\delta x_r}^4$. The autocorrelation decays as $\rho_{\delta x_r}(\tau)|_{\tau \geq 3} = c_\alpha(1 - a_{pd})^\tau + c_\beta \lambda_c^\tau$.

Result. Substituting $\sigma_{\delta x_r}^2 = C_{\text{dpmr}} 4k_B T a_x + C_n \sigma_n^2$ gives $\sigma_{\delta x_r}^4 / (C_{\text{dpmr}} 4k_B T)^2 = (a_x + \xi)^2$, hence

$$R(2, 2) = K_{\text{var}} \text{IF} (a_x + \xi)^2, \quad \xi = \frac{C_n}{C_{\text{dpmr}}} \cdot \frac{\sigma_n^2}{4k_B T}.$$

The relative noise $\sigma(a_{xm})/a_x \approx \sqrt{K_{\text{var}} \text{IF}}$ (thermal-dominated, $\xi \ll a_x$) is an irreducible finite-sample (chi-squared) floor, set by the EWMA window $N_{\text{eff}} \approx 1/a_{\text{cov}}$.

Delay term and KF measurement

Gain model (4-state; the gain rate δa_x is fed forward from the desired trajectory, δa_{ram} is the random part):

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k],$$

$$\delta a_x[k] = a'_x[k] \Delta h_d[k] \text{ (known, fed forward)}, \quad \delta a_{ram}[k] = a'_x[k] \Delta x_{ram}[k].$$

The gain measurement reflects the gain d steps earlier; expanding $a_x[k-d]$ and separating the known fed-forward part,

$$y_2[k] = a_{xm}[k] = a_x[k-d] + n_a[k] = a_x[k] - \underbrace{\sum_{i=1}^d \delta a_x[k-i]}_{r_2[k]} + \left\{ n_a[k] - \sum_{i=1}^d \delta a_{ram}[k-i] \right\}.$$

Since $\sum_{i=1}^d \delta a_x[k-i]$ is known, it is moved to a known correction $\mathbf{c}[k]$, and the state keeps only a_x :

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \delta a_x[k-i] \end{bmatrix}, \quad \mathbf{y}[k] = \mathbf{H} \mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k].$$

Effective measurement variance (the intrinsic IIR noise n_a and the past random gain increments are independent):

$$R(2, 2)_{\text{eff}} = \underbrace{R(2, 2)}_{=\text{Var}(n_a)} + \sum_{i=1}^d \text{Var}(\delta a_{ram}[k-i]), \quad \text{Var}(\delta a_{ram}) = \frac{2}{1 + \lambda_c} \left(\frac{a_x K_h}{R} \right)^2 \sigma_{\delta h}^2,$$

with $K_h = (1/c)(dc/d\bar{h})$ the wall sensitivity, R the probe radius, and $\sigma_{\delta h}^2 = 4k_B T a_{\perp}$ the wall-normal thermal step variance. The random-gain term $\delta x_{\delta af}^d$ (the contribution of δa_{ram} to δx) is second order and is omitted from the C_{dpmr} derivation above.

Complete signal path: $p_{xm} \rightarrow a_{xm}$

$$\delta x_m[k] = p_{xd}[k - d] - p_{xm}[k], \quad d = 2 \quad (1)$$

$$\delta x_{md}[k] = (1 - a_{pd}) \delta x_{md}[k - 1] + a_{pd} \delta x_m[k] \quad (2)$$

$$\delta x_r[k] = \delta x_m[k] - \delta x_{md}[k] \quad (3)$$

$$\hat{\sigma}_{\delta x_r}^2[k] = (1 - a_{cov}) \hat{\sigma}_{\delta x_r}^2[k - 1] + a_{cov} \delta x_r^2[k] \quad (4)$$

$$\boxed{a_{xm}[k] = \frac{\hat{\sigma}_{\delta x_r}^2[k] - C_n \sigma_n^2}{C_{\text{dpmr}} \cdot 4k_B T}} \quad (5)$$